

Concept of Economics in Engineering

- • Economics is the study of how people allocate scarce resources.
- • In engineering, economics helps in decision-making regarding cost, efficiency, and resource use.
- • Engineers use economic principles to evaluate the feasibility and profitability of projects.
- • It includes concepts like cost-benefit analysis, value engineering, and optimization.

Scope of Economics in Engineering

- • Project Evaluation and Investment Analysis
- • Cost Estimation and Control
- • Budgeting and Financial Planning
- • Resource Allocation and Optimization
- • Economic Feasibility Studies
- • Decision-Making under Uncertainty

Microeconomics in Engineering Context

- • Focuses on individual units like firms, industries, and consumers.
- • Engineers apply microeconomics to optimize production, pricing, and resource allocation.
- • Helps in analyzing market structures and firm behavior.
- • Relevant for cost analysis, break-even analysis, and pricing strategies.

Macroeconomics in Engineering Context

- • Deals with the economy as a whole: GDP, inflation, unemployment, etc.
- • Helps engineers understand economic trends that affect industry and investments.
- • Useful for strategic planning and forecasting.
- • Affects decisions related to infrastructure, energy, and technology investments.

Why Economics is Important for Engineers

- • Enables efficient use of resources in projects.
- • Supports cost-effective and sustainable solutions.
- • Enhances decision-making and risk assessment.
- • Essential for project management and feasibility studies.

Role of Economics in Engineering Decisions

- • Selection of technology and materials
- • Design and process optimization
- • Investment and financing decisions
- • Lifecycle cost analysis
- • Market and demand analysis

- Key Economic Problems

Scarcity: The Fundamental Economic Problem

- • Scarcity means limited resources to meet unlimited wants.
- • Engineers face scarcity in terms of budget, materials, time, and manpower.
- • It requires prioritization and trade-offs in decision-making.
- • Economic analysis helps manage these constraints efficiently.

Resource Allocation in Engineering

- • Refers to the distribution of limited resources among competing projects or activities.
- • Goal is to maximize efficiency and output.
- • Engineers use tools like linear programming, cost-benefit analysis, and project evaluation.
- • Sound allocation leads to optimal project outcomes.